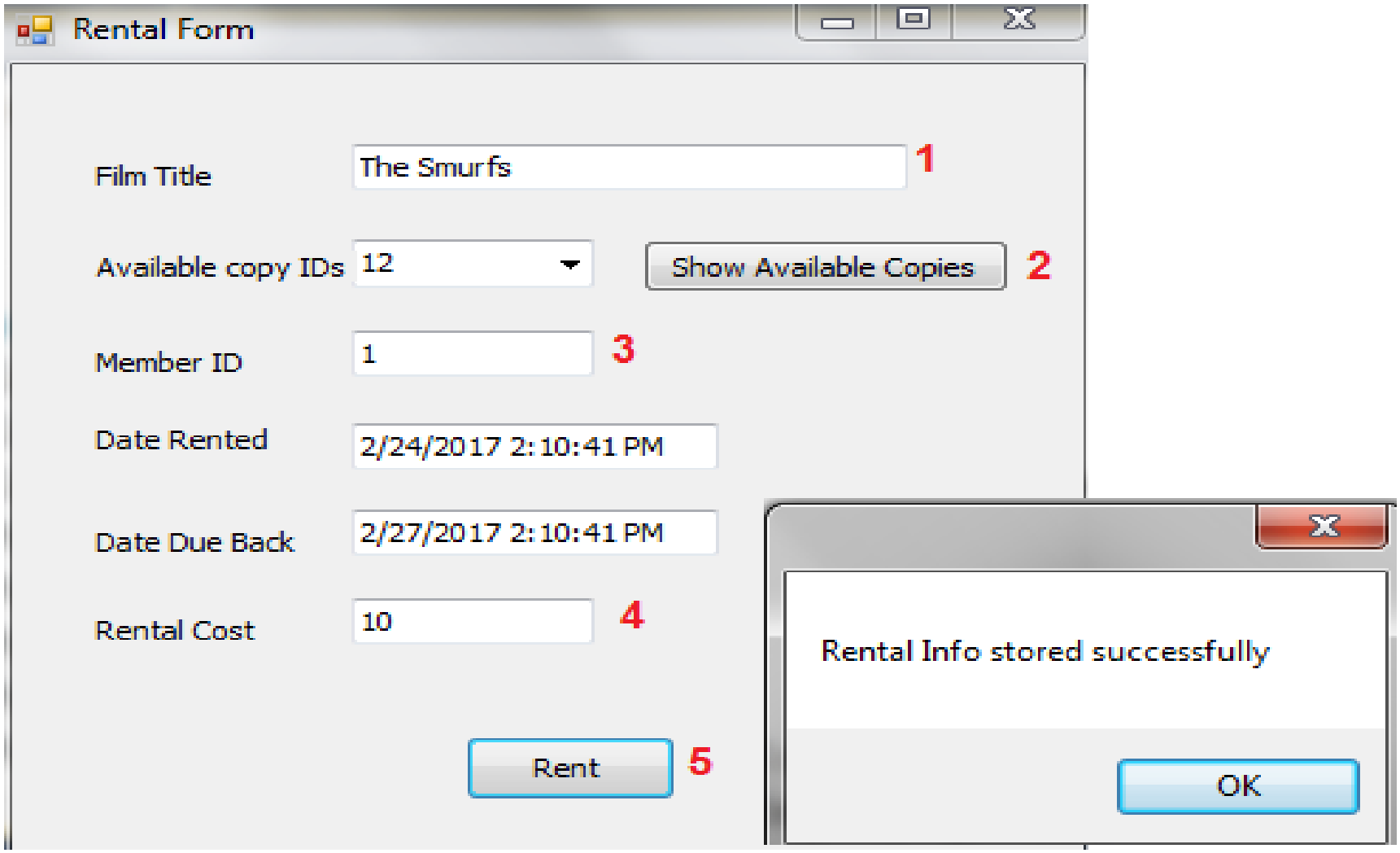
Software Engineering Lab

Building .NET Applications on the Oracle Database with Microsoft Visual Studio

Lab 4 – ODP.Net
Oracle Stored Procedures connected layer

Rental form


Rental Form
Film Title
The Smurfs
1
Available copy IDs
12
Show Available Copies
2
Member ID
1
3
Date Rented
2/24/2017 2:10:41 PM
Date Due Back
2/27/2017 2:10:41 PM
Rental Cost
10
4
Rent
5
Rental Info stored successfully
OK

Implementation Steps

1- Loading FilmCopies ComboBox with IDs of Available copies in showCopies_click event

2- Set DateRented with today's date

3- Set Due Date with three days from the date the DVD is rented

4- To Insert a new rental, we must pass:

- RentID, MemberID, FilmCopyID, DateRented, Due Date, Rental Cost to Rental Table

- So we must store New value for RentID

5- Set the film status to RENTED

4- Insert New Rental Information

- Procedure "GetRentID" to get Max RentID from the rentals table
  - After returning the Max RentID, its value will be incremented by 1 and new value will be used as new rentID

- Procedure "Insert_Rental" to insert new record in Rentals table
  - We must pass(RentID, MemberID, FilmCopyID, DateRented, Due Date, Rental Cost) To the insertion procedure.

4- Insert New Rental Information
A. Create Proc "GetRentID"

A. Procedure "GetRentID" to get Max RentID from the rentals table

```
create or replace procedure GetRentID (RID out number)
as
begin
select max(rentid)
into RID
from rentals;
end;
```

4- Insert New Rental Information
A. calling "GetRentID"

```
private void btn_rent_Click(object sender, EventArgs e)
{
    int maxID , newID;
    OracleCommand cmd = new OracleCommand();
    cmd.Connection = con;
    cmd.CommandText = "GetRentID";
    cmd.CommandType = CommandType.StoredProcedure;
    cmd.Parameters.Add("id", OracleDbType.Int32, ParameterDirection.Output);
    cmd.ExecuteNonQuery();
    try
    {
        maxID = Convert.ToInt32(cmd.Parameters["id"].Value.ToString());
        newID = maxID + 1;
    }
    catch
    { newID = 1; }

}
```

4- Insert New Rental Information
A. calling "GetRentID"

- Syntax to read value from output parameter:

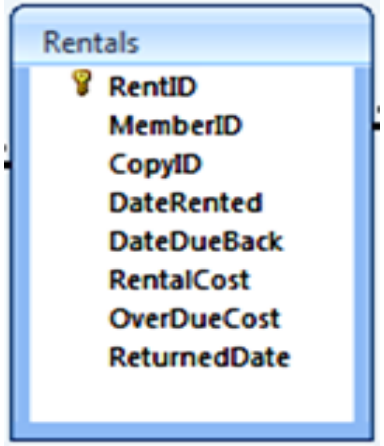
```
data_type variable_name=
cmd.parameters["parameter_name"].Value.ToString())
```

  - If the returned value is null, it will throw and exception. To avoid exceptions, write the above statement between try and catch.
  - It returns Object data type, we must cast the returned value to the required data type using convert.To (...) function

4- Insert New Rental Information
B. Create Proc "Insert_rental"

B. Procedure "Insert_Rental" to insert new record in Rentals table

```
create or replace procedure Insert_Rental
(RID number, MID Number,CID Number,
DRented  Date, DBack  Date, Cost  Number)
as
begin
insert  into  Rentals
(RentID, MemberID,CopyID, DateRented, DateDueBack, RentalCost)
Values ( RID, MID, CID, DRented, DBack, Cost) ;
End ;
```

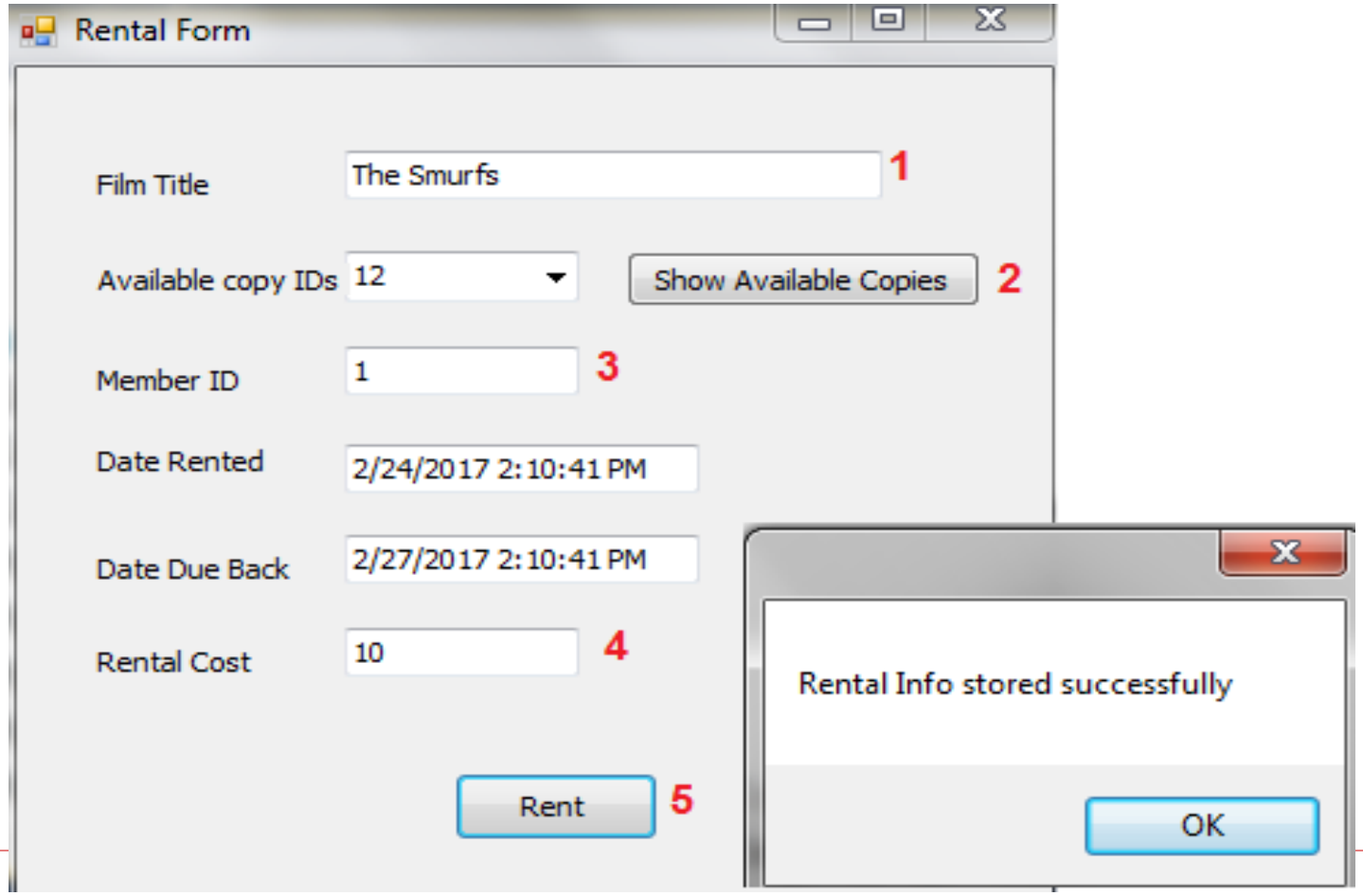


4- Insert New Rental Information
B. calling "Insert_rental"

- Calling The Procedure From .Net:
  - Continue in the btn_Rent_click event

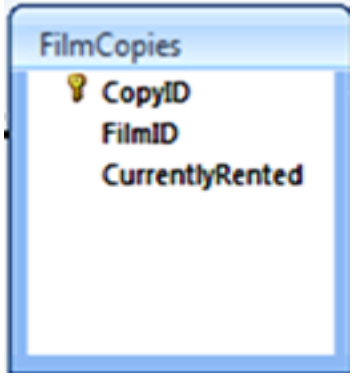
```
OracleCommand c = new OracleCommand();
c.Connection = con;
c.CommandText = "Insert_Rental";
c.CommandType = CommandType.StoredProcedure;
c.Parameters.Add("RID", newID);
c.Parameters.Add("MID", txt_memID.Text);
c.Parameters.Add("CID", cmb_copies.SelectedItem.ToString());
c.Parameters.Add("D1", Convert.ToDateTime(txt_DateRented.Text));
c.Parameters.Add("D2", Convert.ToDateTime(txt_DateBack.Text));
c.Parameters.Add("cost", txt_cost.Text);
c.ExecuteNonQuery();
MessageBox.Show("Rental Info stored successfully ");
```

4- Insert New Rental Information



5- Set the film status to RENTED
Create Proc. Update_status

```
create or replace procedure update_status
  (CID Number)
as
begin
     update    FilmCopies
     set   CurrentlyRented  =   'y'
     where   CopyID  =   CID  ;
end;
```



5- Set the film status to RENTED

- Calling The Procedure From .Net:
  - Continue in the btn_Rent_click event

```
OracleCommand c2 = new OracleCommand();
c2.Connection = con;
c2.CommandText = "update_status";
c2.CommandType = CommandType.StoredProcedure;
c2.Parameters.Add("id", cmb_copies.SelectedItem.ToString());
c2.ExecuteNonQuery();
MessageBox.Show("Film status updated succefully");
```

Thank You